

Concentration of a strong electrolyte (NaCl), K_a of a weak electrolyte (CH_3COOH) from conductivity measurements

Aim

Classify a compound weak and strong electrolyte from the conductivity measurement

Find out how conductivity of the solution changes as a function of concentration for a weak/strong electrolyte

Determination of Concentration of unknown solution of a strong electrolyte

Determination of K_a of a weak electrolyte (acetic acid)

The conductivity of a solution is proportional to the number of ions in unit volume and , therefore, can be used to determine the concentration of an electrolyte solution. According to Kohlrausch's law, the conductivity of a solution of a strong electrolyte (at low concentrations, $c < 10^{-2}$ M) is related to concentration,

$$\Lambda = \Lambda_0 - K\sqrt{c} \quad (1)$$

where Λ is molar conductivity in $\text{S cm}^2 \text{mol}^{-1}$, Λ_0 is the limiting molar conductivity in $\text{S cm}^2 \text{mol}^{-1}$, K is a non-negative constant in $\text{S cm}^2 \text{mol}^{-1} \text{M}^{-1/2}$, and c is the molar concentration in M. Molar conductivity is defined as the measured conductivity of an electrolyte solution (κ in mS cm^{-1}) divided by the molar concentration of the electrolyte (c)

$$\Lambda = \frac{\kappa}{c} \quad (2)$$

Concentration of a strong electrolyte



Take stock solution of 0.1 M NaCl.



Make 0.05 M, 0.02 M NaCl, 0.01 and 0.005 M NaCl solution



Make a calibration curve of molar conductivity versus square root of concentration



Calculate the concentration of the unknown solution

Determination of K_a of a weak electrolyte (acetic acid)

Know your Chemical by Sign



Flammable



Oxidizer



Explosive



Low Level Hazard



Corrosive



***Severe Chronic
Hazard***



Poison



***Environmental
Hazard***

MSDS of Acetic Acid



Flammable liquids
Skin Corrosion/Irritation
Serious Eye Damage/Eye Irritation

Category 3
Category 1 A
Category 1

Hazard Statements

Flammable liquid and vapor

Causes severe skin burns and eye damage

Keep away from heat/sparks/open flames/hot surfaces. - No smoking

Take precautionary measures against static discharge

Wear protective gloves/protective clothing/eye protection/face protection

Do not breathe dust/fume/gas/mist/vapors/spray

Use only outdoors or in a well-ventilated area

Wash face, hands and any exposed skin thoroughly after handling



Response

Immediately call a POISON CENTER or doctor/physician

Inhalation

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

Skin

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

Wash contaminated clothing before reuse

Eyes

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Ingestion

IF SWALLOWED: Rinse mouth. DO NOT induce vomiting



Determination of K_a or K_b values from pH measurements is well-known, hence will not be described here. However, determination of K_a from conductivity measurements is less well-known and therefore a description is provided. The molar conductivity of a weak acid (Λ_M) is given by the expression

$$\Lambda_M = \frac{\Lambda}{c} \quad (1)$$

where Λ is the conductivity of the weak acid and c is its molarity. The fraction (α) dissociated is

$$\alpha = \frac{\Lambda_M}{\Lambda_o} \quad (2)$$

where Λ_o is the molar conductivity at infinite dilution. For strong electrolytes, Λ_o can be determined by plotting Λ_M versus square root of the concentration; Λ_o is the y intercept of this linear plot. For weak electrolytes, a plot of Λ_M versus square root of the concentration curves and so Λ_o is obtained indirectly. For example, for acetic acid, $\Lambda_o(\text{HC}_2\text{H}_3\text{O}_2)$ is

$$\Lambda_o(\text{HC}_2\text{H}_3\text{O}_2) = \Lambda_o(\text{NaC}_2\text{H}_3\text{O}_2) + \Lambda_o(\text{HCl}) - \Lambda_o(\text{NaCl})$$

Table 1. Method 1 Conductivity Data and Results at $\sim 20^\circ\text{C}$

$V(1.0\text{ M HC}_2\text{H}_3\text{O}_2)/$ mL	$[\text{HC}_2\text{H}_3\text{O}_2]_0/$ (mol/L)	Conductivity/ ($\mu\text{S}/\text{cm}$)	Molar Conductivity/ ($\mu\text{S}/\text{cm M}$)	Fraction Ionized (α)	$[\text{H}^+] =$ $[\text{C}_2\text{H}_3\text{O}_2^-]/$ (mol/L)	$[\text{HC}_2\text{H}_3\text{O}_2]/$ (mol/L)	K_a
1	0.0163	188	11506	2.91×10^{-2}	4.75×10^{-4}	1.59×10^{-2}	1.42×10^{-5}
2	0.0322	258	8024	2.03×10^{-2}	6.52×10^{-4}	3.15×10^{-2}	1.35×10^{-5}
3	0.0475	324	6826	1.73×10^{-2}	8.19×10^{-4}	4.66×10^{-2}	1.44×10^{-5}
4	0.0623	385	6179	1.56×10^{-2}	9.73×10^{-4}	6.13×10^{-2}	1.55×10^{-5}
5	0.0767	422	5503	1.39×10^{-2}	1.07×10^{-3}	7.56×10^{-2}	1.51×10^{-5}
6	0.0906	452	4987	1.26×10^{-2}	1.14×10^{-3}	8.95×10^{-2}	1.46×10^{-5}
7	0.1042	488	4685	1.18×10^{-2}	1.23×10^{-3}	1.03×10^{-1}	1.48×10^{-5}
Average STD							1.46×10^{-5} 6×10^{-7}

Table 2. Method 2 pH Data and Results at $\sim 20^\circ\text{C}$

$V(1.0\text{ M HC}_2\text{H}_3\text{O}_2)/\text{ mL}$	$[\text{HC}_2\text{H}_3\text{O}_2]_0/(\text{mol/L})$	pH	$[\text{HC}_2\text{H}_3\text{O}_2]/(\text{mol/L})$	K_a
1.00	1.63×10^{-2}	3.32	1.59×10^{-2}	1.44×10^{-5}
2.00	3.22×10^{-2}	3.16	3.15×10^{-2}	1.52×10^{-5}
3.00	4.75×10^{-2}	3.11	4.67×10^{-2}	1.29×10^{-5}
4.00	6.23×10^{-2}	3.02	6.14×10^{-2}	1.49×10^{-5}
5.00	7.67×10^{-2}	2.95	7.56×10^{-2}	1.67×10^{-5}
6.00	9.06×10^{-2}	2.91	8.94×10^{-2}	1.69×10^{-5}
7.00	1.04×10^{-1}	2.89	1.03×10^{-1}	1.61×10^{-5}
Average				1.5×10^{-5}
STD				1×10^{-6}